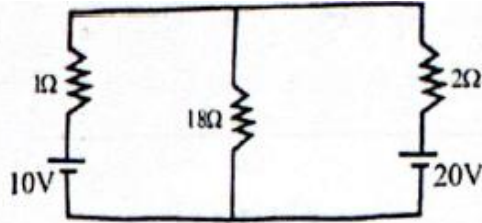


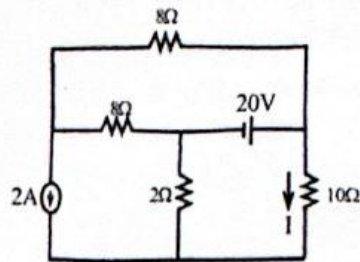
Assigenment: 2

Subject: Basic Electrical and Electronics Engineering.

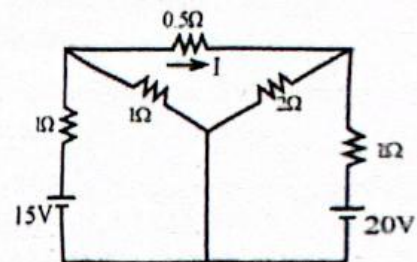
1. Calculate the current through 18Ω resistor of given network using superposition theorem.



2. Determine the value of I in the given network below by using superposition theorem.

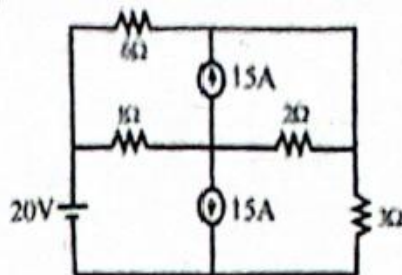


(i)



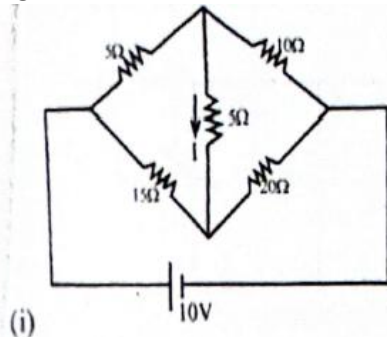
(ii)

3. Calculate the voltage drop and power dissipated in 3Ω resistor using superposition theorem.

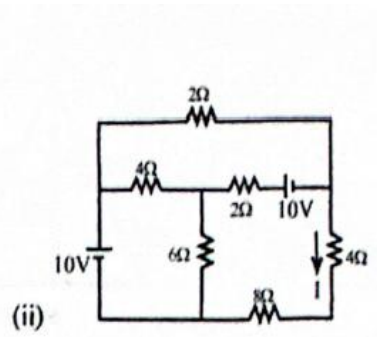


4. Determine the value of I in the network given below

- I. Using Thevenins theorem
- II. Using Nortons theorem.

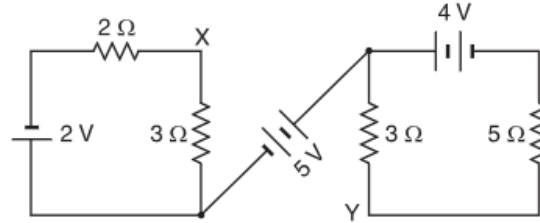


(i)

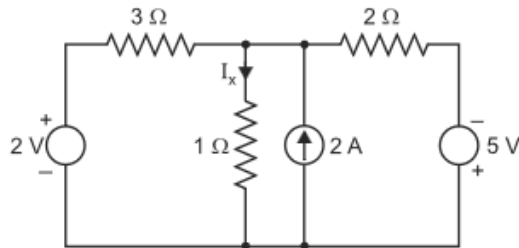


(ii)

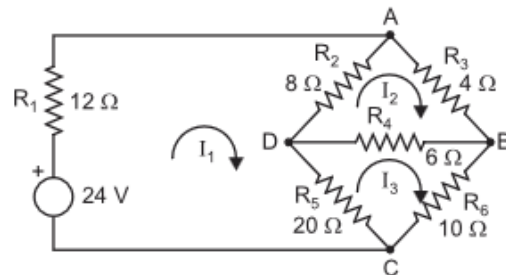
5. What is the potential difference between X and Y in the network shown in Fig.



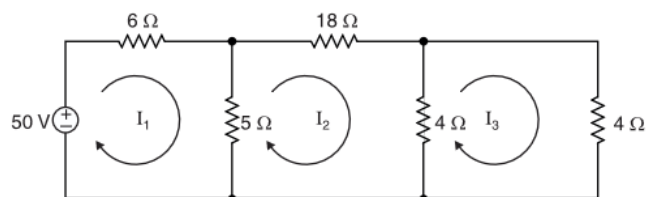
6. Using mesh current method, determine current I_x in the circuit shown in Fig.



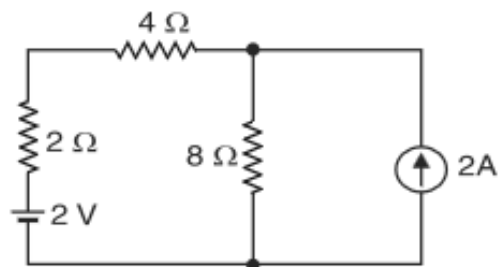
7. Using mesh current method, find the currents in resistances R_3 , R_4 , R_5 and R_6 of the circuit shown.



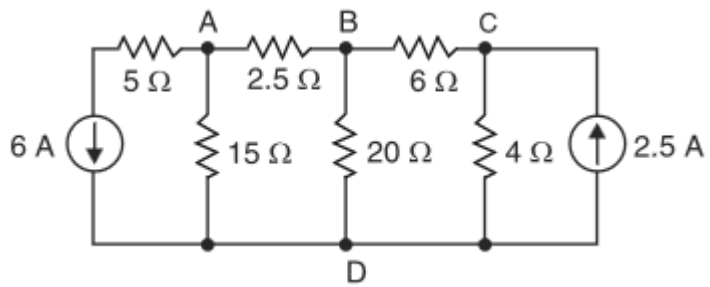
8. For the network shown in Fig find the mesh currents I_1 , I_2 , I_3 .



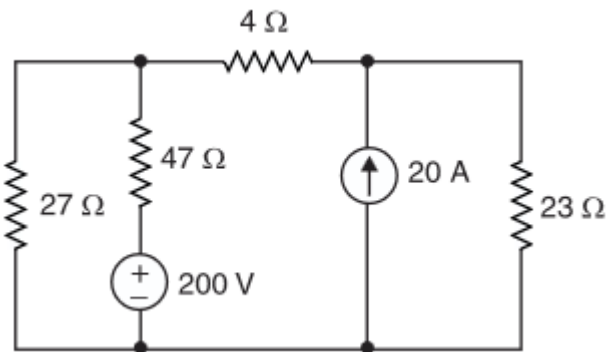
9. Using nodal analysis, find the different branch currents in the circuit shown in figure.



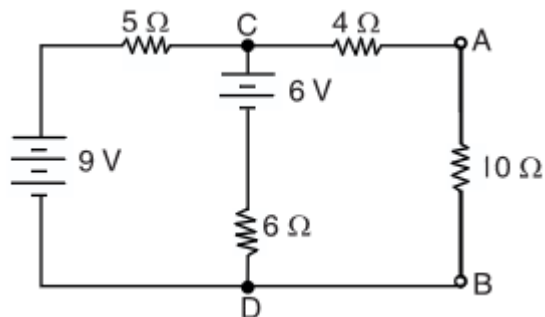
10. Solve the circuit shown in Fig. using nodal analysis.



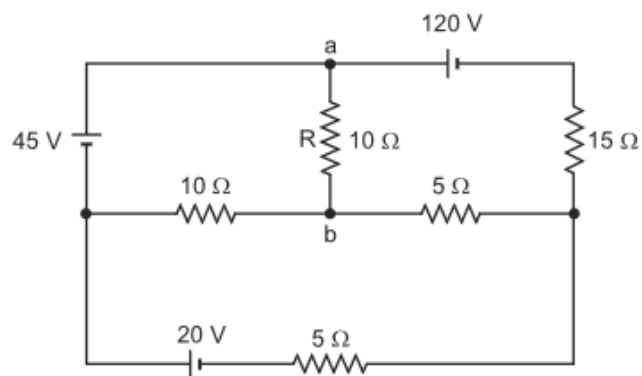
11. Using superposition theorem, find the current in $23\ \Omega$ resistor in the circuit shown below.



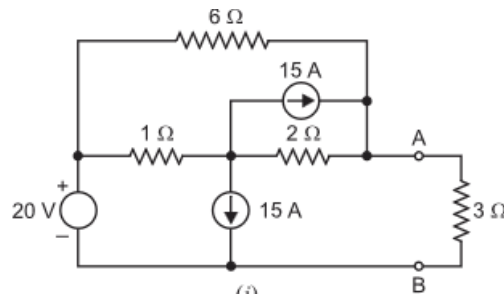
12. Using Thevenin's theorem, find potential difference across terminals AB.



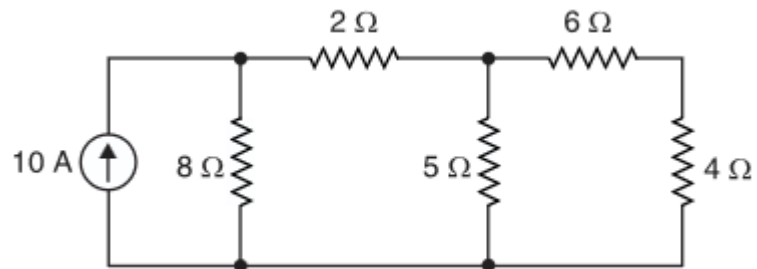
13. Using Thevenin's theorem, find the current through R connected between points a and b.



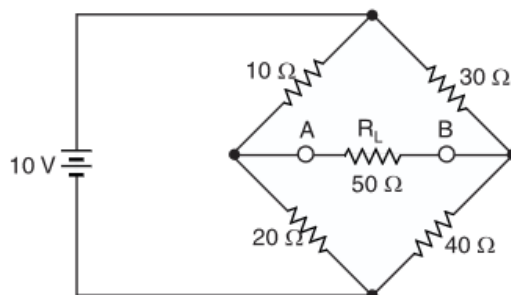
13. Using Thevenin's theorem, find the voltage across 3Ω resistor in Fig.



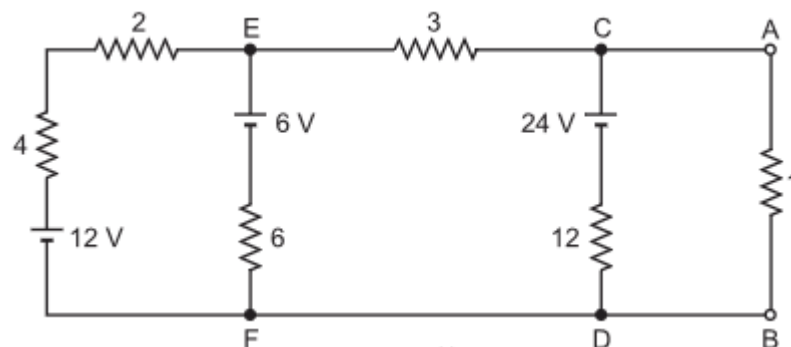
14. Using Norton's theorem, calculate the current in the 5Ω resistor in the circuit.



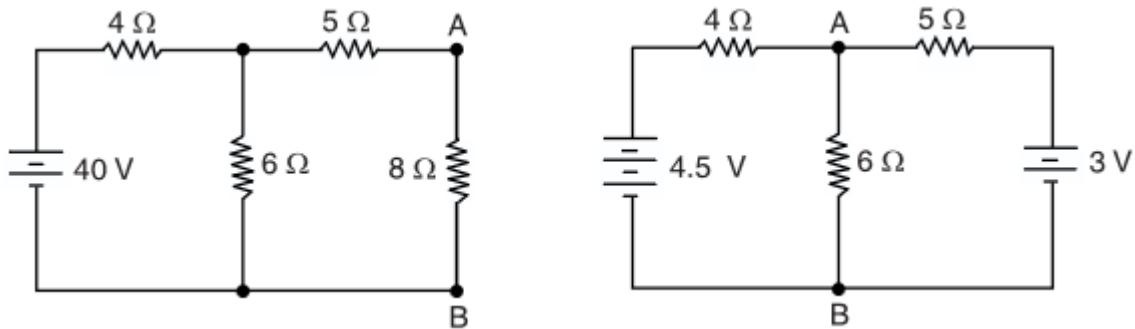
15. Find the Norton current for the unbalanced Wheatstone bridge shown in fig.



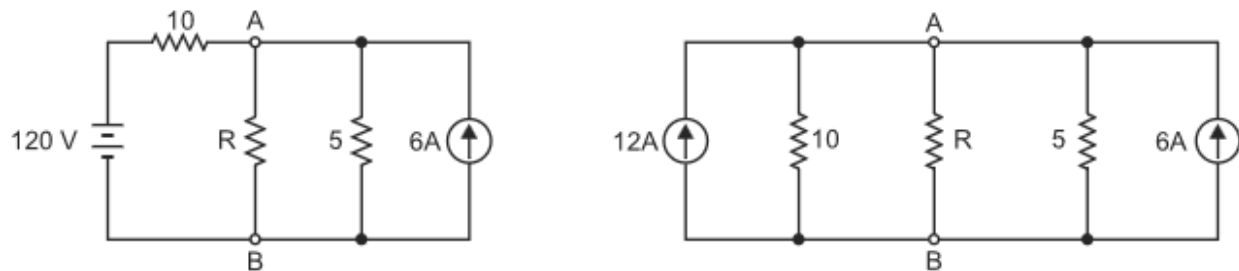
16. Using Norton's theorem, find current through 1Ω resistor in Fig.



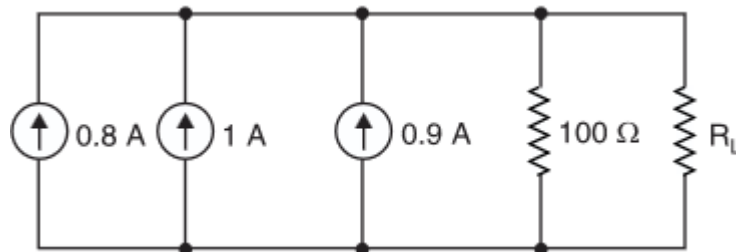
17. Using Norton's theorem, find the current in $8\ \Omega$ resistor of the network shown in Fig.



18. Find the value of resistance R to have maximum power transfer in the circuits shown in Fig. Also obtain the amount of maximum power. All resistances are in ohms.



19. Determine the value of R_L in Fig. for maximum power transfer and evaluate this power.



20. For the circuit find the value of R that will receive maximum power. Determine this power.

